SMS SPAM DETECTION SYSTEM USING NLP

A Project Report

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by

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ABSTRACT

This report details the development and deployment of a machine learning system for SMS spam detection. Leveraging the SMS Spam Collection dataset from Kaggle, containing over 5,500 labelled messages, a classification model was trained to distinguish between spam and legitimate (ham) messages. The project began with a crucial data preprocessing phase, including handling missing/duplicate entries, encoding labels, and performing text cleaning operations such as tokenization, removal of special characters, stop words, and punctuation, and stemming.

Following preprocessing, exploratory data analysis (EDA) was conducted to understand the dataset's characteristics. This involved analysing text length metrics (characters, words, sentences), exploring correlations, and generating word clouds for spam and ham messages. Several machine learning models were evaluated for spam detection, including Naive Bayes, Random Forest, KNN, Decision Tree, Logistic Regression, Extra Trees Classifier, and SVC. Model selection prioritized maximizing precision, with the chosen model achieving a 100% score.

Finally, the trained model was deployed as a user-friendly web application using Streamlit. This application provides a simple interface for users to input SMS messages and receive real-time spam/ham predictions. The project was implemented using Python and its associated libraries, including Scikit-learn, Pandas, NumPy, and Streamlit.

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CHAPTER 1

Introduction

1.1 Problem Statement:

The proliferation of SMS spam creates a significant challenge for mobile phone users. These unwanted messages can expose individuals to threats like phishing and scams, while also disrupting communication by cluttering inboxes and making it difficult to find legitimate messages. The constant influx of spam wastes users' time and can lead to frustration. Manually filtering this influx is inefficient and time-consuming, making an automated solution crucial.

This automated spam detection system offers a solution. By accurately identifying and classifying incoming SMS messages as spam or legitimate, it enhances user security by reducing exposure to threats and improves the messaging experience by filtering out the noise. This allows users to focus on important communications and reclaim control of their inboxes.

Why is this significant?

1. **Security:** Protects users from phishing and scams.
2. **Efficiency:** Automates spam filtering, saving time.
3. **Communication:** Improves message clarity by reducing clutter.
4. **User Experience:** Enhances overall messaging experience.

1.2 Motivation:

The increasing prevalence of phishing and scam messages makes manual filtering an inefficient and time-consuming burden. Artificial intelligence offers a solution by classifying messages more accurately, thus reducing the risk of users falling victim to fraudulent schemes. This automated approach minimizes spam, improves communication flow, and enhances the overall messaging experience.

Potential Applications and Impact:

1. **Enhanced Mobile Security:** Directly protects users from SMS-based threats.
2. **Improved Communication Efficiency:** Streamlines messaging by filtering out clutter.
3. **Integration with Messaging Platforms**: Can be seamlessly incorporated into existing SMS apps.
4. **Reduced User Frustration:** Provides a more positive and secure messaging experience.

1.3 Objective:

The goal is to develop and evaluate a robust SMS spam detection system using machine learning and deep learning techniques.

Specific Objectives:

1 Implement and compare the performance of various machine learning models (e.g., Naive Bayes, Random Forest, SVM) for SMS spam detection.

2 Explore and evaluate deep learning models (e.g., RNNs, CNNs) for enhanced spam detection capabilities.

3 Assess the performance of all models using standard evaluation metrics such as precision, recall, F1-score, and accuracy.

1.4 Scope of the Project:

Scope:

The system focuses on developing an SMS spam detection system trained using a labeled dataset and employing supervised learning models. The system aims to accurately classify messages as either spam or ham.

1. Builds a spam detection model using supervised learning.
2. Trains the model with a labeled SMS spam dataset.
3. Compares different machine learning algorithms.
4. Measures performance using standard metrics.
5. Classifies messages as spam or ham.

Limitations:

This section acknowledges the inherent constraints of the project, outlining the factors that may impact the system's performance and the areas that are beyond the current project's capabilities. It addresses potential weaknesses, dependencies, and areas for future improvement.

1. Model accuracy is dependent on the quality and representativeness of the training data.
2. The model may struggle with ambiguous or cleverly disguised spam messages.
3. Spammers can adapt their tactics to evade detection.
4. Handling SMS data involves inherent security and privacy risks.
5. The current system may lack real-time detection capabilities and multi-lingual support.

CHAPTER 2

Literature Survey

2.1 Review relevant literature

The evolution of SMS spam detection techniques, from simple keyword filtering to sophisticated deep learning models, is explored in this literature review, highlighting the key advancements and challenges in the field.

1. **Rule-Based and Early Machine Learning Approaches**

   SMS spam detection has evolved from simple rule-based systems, easily circumvented by spammers, to more sophisticated machine learning approaches. Early machine learning models, like Naïve Bayes and SVMs, offered improved performance by leveraging statistical properties of text. These models, as explored by authors like Cormack et al. (2007) with SVMs, could learn from labeled data and generalize to unseen messages. The creation of benchmark datasets, like the SMS Spam Collection by Almeida and Hidalgo (2011), facilitated comparison and evaluation.

2. **Deep Learning for Contextual Understanding**

   Deep learning has further revolutionized the field. Models like Recurrent Neural Networks (RNNs), as investigated by Kim et al. (2017), and Convolutional Neural Networks (CNNs) have demonstrated the ability to capture complex patterns and contextual information in text. These models can learn intricate relationships between words and phrases, improving spam detection.

3. **Feature Engineering and Word Embeddings**

   Effective feature engineering is crucial. Techniques like TF-IDF and Word2Vec convert text into numerical representations. TF-IDF weighs word importance, while Word2Vec captures semantic relationships. Chen et al. (2015) explored various feature engineering strategies, highlighting their importance.

4. **Transformer Models and the Rise of Contextualized Embeddings**

   Transformer-based models, particularly BERT, have achieved state-of-the-art results. The work by Vaswani et al. (2017) and the development of BERT by Devlin et al. (2018) have enabled models to better understand context and linguistic relationships. BERT's bidirectional training captures nuanced language patterns, improving spam detection accuracy.

5. **Ongoing Challenges and Future Directions**

   While progress has been made, developing robust and adaptive systems remains a challenge. Spammers adapt their tactics, requiring models that generalize well. Future research includes exploring novel deep learning architectures and developing techniques to counter adversarial attacks. Data privacy and security are also critical areas of focus.

2.2 Existing Models, Techniques, and Methodologies

1. **Rule-based systems:** Simple to implement but easily bypassed by spammers; rely on predefined keywords or patterns.
2. **Machine learning:** Learns from data, offering better generalization than rule-based systems; various algorithms with different strengths and weaknesses.
3. **Deep learning:** Captures complex patterns and context, often achieving higher accuracy; requires more data and computational resources.
4. **Feature engineering:** Crucial for representing text data effectively; transforms text into numerical features suitable for models.
5. **Evaluation**: Quantifies model performance; different metrics emphasize different aspects of classification accuracy.

2.3 Limitations in Existing Systems

1. **Evolving Spam Tactics:** Spammers constantly adapt their methods, using new keywords, obfuscation techniques, and social engineering tactics. Models trained on past spam patterns may struggle with these novel approaches, requiring continuous retraining and adaptation.
2. **Contextual Understanding:** Even advanced models struggle with subtle linguistic nuances like sarcasm, irony, or implied meaning. Accurately capturing the intent and meaning behind a message, especially in short-form text like SMS, remains a significant challenge.
3. **Data Dependence:** Model performance heavily relies on the quality and quantity of training data. Biased or limited datasets can lead to poor generalization and reduced accuracy on unseen data, highlighting the need for large, representative datasets.
4. **Computational Cost:** Deep learning models, particularly Transformer-based models, can be computationally expensive to train and deploy, limiting their applicability in resource-constrained environments like mobile devices. This cost can be a barrier to real-time spam detection on mobile phones.

How This Project Addresses the Gaps

This project aims to address key limitations in existing SMS spam detection systems by:

1. **Leveraging BERT for Enhanced Contextual Understanding:** Addressing the challenge of accurately interpreting nuanced language, including sarcasm and implied meaning, by utilizing BERT's powerful contextualized word embeddings and bidirectional training, which captures deeper semantic relationships within SMS messages.

2. **Developing a Dynamic Feature Set:** Tackling the issue of evolving spam tactics by incorporating a dynamic feature set that includes not only traditional features like TF-IDF and word embeddings but also features designed to capture evolving spam patterns, such as the presence of shortened URLs, unusual character combinations, and sentiment shifts, allowing the model to adapt to new spam techniques.

3. **Focusing on Targeted Spam Detection:** Improving the detection of specific types of spam, such as phishing and smishing attacks, by creating specialized models trained on datasets enriched with examples of these targeted attacks. This approach allows for the development of more accurate and effective detection strategies for these high-risk spam categories.

4. **Rigorous Evaluation on a Diverse Dataset:** Mitigating the risk of data dependence and ensuring model generalizability by evaluating the model's performance on a large and diverse dataset that includes various spam types, languages (if applicable), and real-world SMS messages collected from different sources. This comprehensive evaluation helps to assess the model's robustness and its ability to perform well in real-world scenarios.

CHAPTER 3

Proposed Methodology

The proposed methodology presents a comprehensive approach to building an SMS spam detection system, covering data preprocessing, exploratory analysis, model training, and web deployment for real-time spam classification.

3.1 System Design

The system architecture follows a pipeline approach, starting with data acquisition and culminating in a deployed web application as shown in Fig. 1.

1. **Data Collection:** The system begins with the collection of SMS messages from the SMS Spam Collection dataset available on Kaggle. This dataset contains over 5,500 SMS messages labeled as either 'spam' or 'ham' and serves as the foundation for training and evaluating the spam detection model.

2. **Data Preprocessing:** The collected data undergoes a rigorous preprocessing stage using Python with the Pandas and NumPy libraries. This involves:

   Cleaning: Handling null values and removing duplicate messages. The 'type' column, indicating spam or ham, is label-encoded. Text normalization is performed by converting all text to lowercase.

   Tokenization: Splitting the text into individual words or tokens.

   Stop Word Removal: Eliminating common words (e.g., 'the,' 'a,' 'is') that do not contribute significantly to spam detection.

   Punctuation Removal: Removing punctuation marks and special characters.

   Stemming: Reducing words to their root form to improve model generalization.

3. **Exploratory Data Analysis (EDA):** EDA is performed using Python with the Pandas and Matplotlib (pyplot) libraries to gain insights into the characteristics of the data. This includes:

   Calculating descriptive statistics for message length (character, word, and sentence count).

   Visualizing data distributions using histograms, bar charts, pie charts, and box plots (5-number summaries).

   Generating heatmaps to analyse correlations between variables.

Creating word clouds to identify frequently occurring terms in spam and ham messages. The most frequent words in spam texts were also visualized separately.

4. **Model Building and Selection:** Several machine learning models from the Scikit-learn library are trained and evaluated, including Naive Bayes, Random Forest, KNN, Decision Tree, Logistic Regression, ExtraTreesClassifier, and SVC. The models are trained on the preprocessed SMS data, and their performance is assessed using metrics such as precision, recall, F1-score, and accuracy. The best classifier was chosen based on achieving 100% precision.

5. **Web Deployment:** The trained and selected model is deployed as a web application using Streamlit. This allows users to input SMS messages through a simple input box and receive real-time spam predictions. The web application provides a user-friendly interface for interacting with the model.

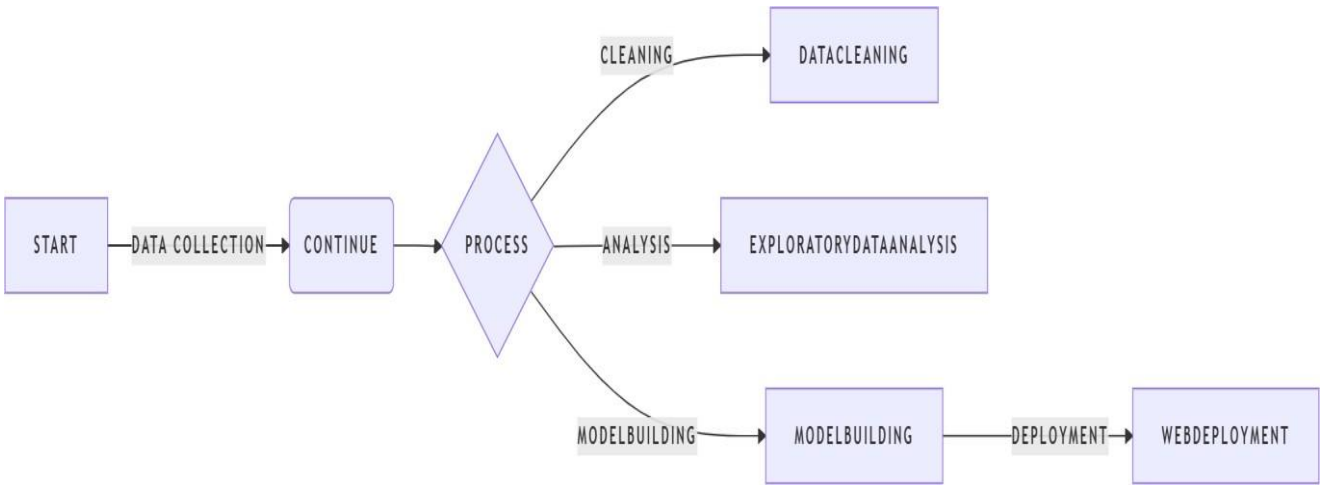

Figure 1: System Workflow for SMS Spam Detection System Using NLP System.

3.2 Requirement Specification

3.2.1 Hardware Requirements:

1. **Development PC:** Standard desktop or laptop for coding, data processing, and model training.
2. **Storage:** Hard drive or SSD for data and code.
3. **Network:** Internet connection for data download and deployment.
4. **Deployment Server:** Chosen model was deployed as a web application using Streamlit.

3.2.2 Software Requirements:

1. **Programming Language:** Python (including libraries like Pandas, NumPy, Scikit-learn, and Matplotlib)
2. **Data Processing and Analysis Libraries:**

   Pandas: For data manipulation and analysis.

   NumPy: For numerical computing.
3. **Machine Learning Library:** Scikit-learn: For model building, selection, and evaluation.
4. **Data Visualization Library:** Matplotlib (specifically pyplot): For creating visualizations.
5. **Web Framework:** Streamlit: For creating and deploying the web application.
6. **Development Environment:** A suitable IDE or text editor (e.g., VS Code, PyCharm, Jupyter Notebooks).
7. **Operating System:** Any OS that supports Python and the required libraries (Windows, macOS, Linux). Deployment environments might have specific OS requirements.
8. **Web Browser:** For accessing the Streamlit web application.

CHAPTER 4

Implementation and Result

In this section, we describe the implementation process and analyze the results of the SMS spam detection system. We leverage visualizations such as a pie chart to understand class balance, a histogram to explore message length characteristics, and a word frequency bar chart to gain insights into common spam vocabulary, which inform both feature engineering and model evaluation.

4.1 Snap Shots of Result:

This pie chart illustrates the distribution of spam and non-spam SMS messages in the dataset. The clear majority of messages (87.37%) are classified as not spam, 12.63% are identified as spam.

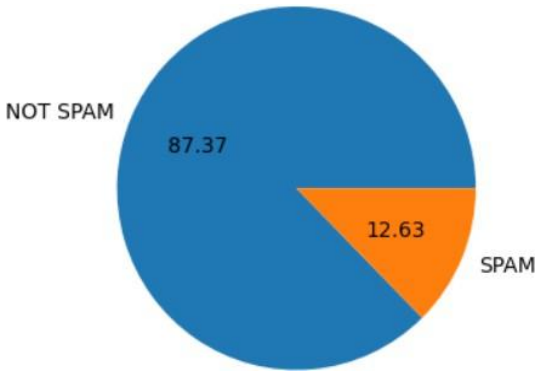

Fig.2 Pie chart for the Spam Detection System.

The histogram visualizes the frequency of different character counts in SMS messages. The yellow bars represent the 'ham' or non-spam class, predominantly found in the lower character count ranges. The darker bars, representing 'spam' messages, demonstrate a broader spread across higher character counts, indicating that spam messages may often be longer. The axes represent the count of messages (y-axis) and the number of characters in each message (x-axis) as shown in Fig 2.

The bar chart visualizes the top 50 most frequent words observed in the spam portion of the SMS dataset. The vertical axis shows the frequency count, while the horizontal axis lists the individual words. The data reveals a tendency for spam messages to contain words associated with promotional offers ('free,' 'prize'), communication ('call,' 'text'), and urgency ('urgent'), among others. This information can be valuable in distinguishing spam from legitimate messages as shown as Fig 3.

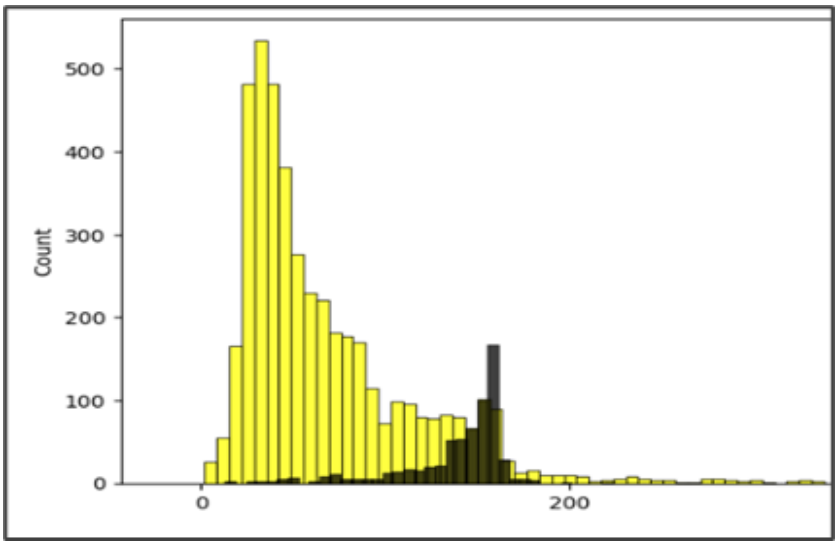

Figure 3: Distribution of Character Counts in SMS Messages
(Spam vs. Non-Spam)

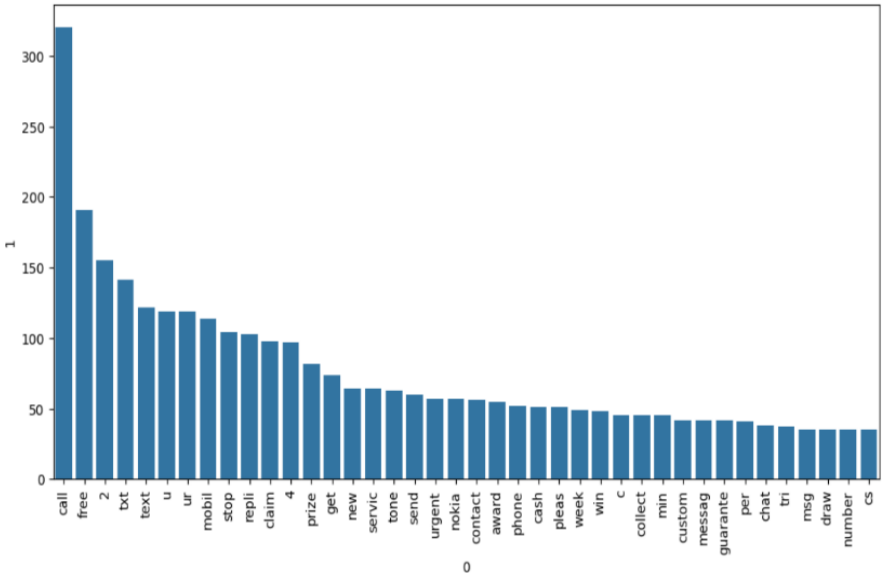

Figure 4 : Bar chart word Frequency Analysis: Top 50 Spam Terms

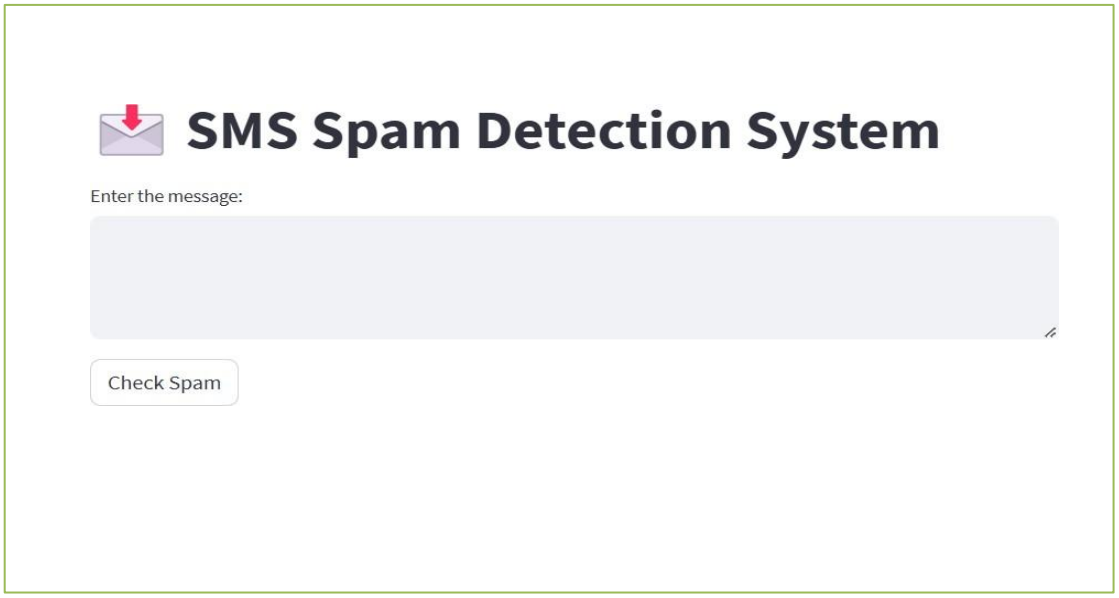

Figure 5 : Interface of the Spam Detection Model

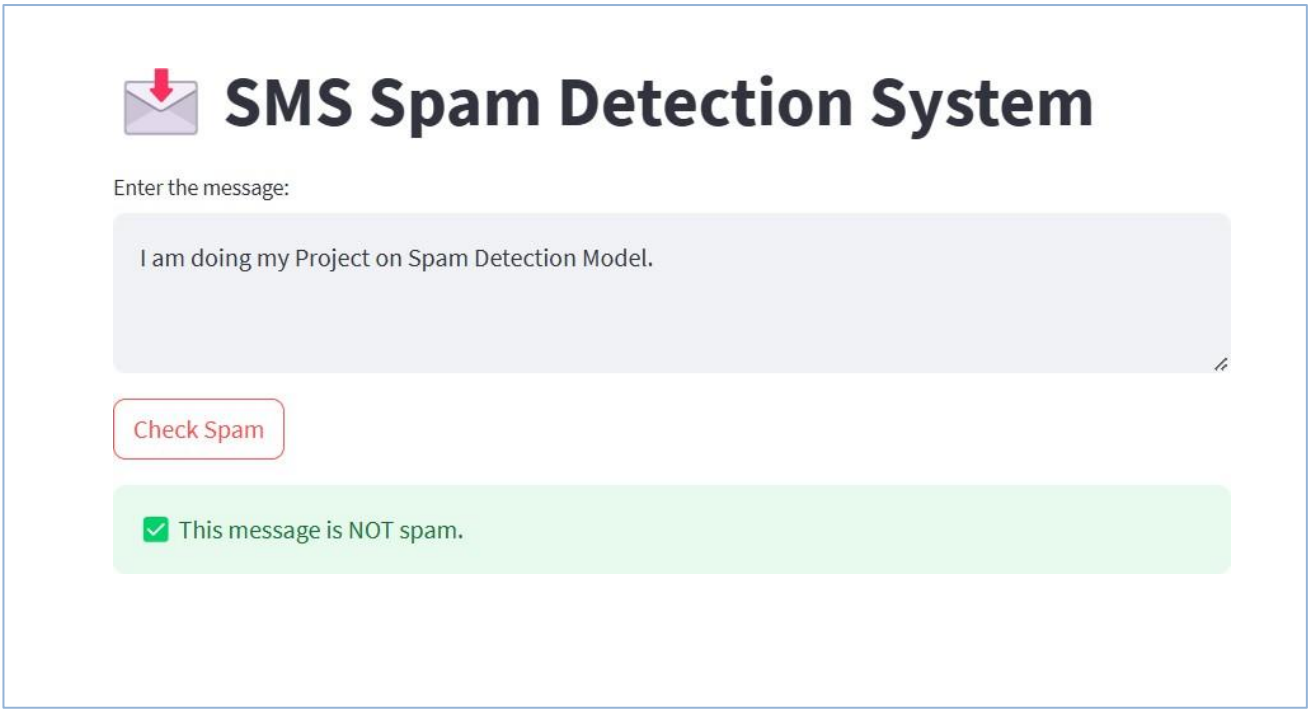

Figure 6: Checking if the message entered is Spam or not spam.

This image shows the interface of an SMS Spam Detection System. A user input message, "I am doing my Internship under Edunet Foundation," was classified as "NOT spam" by the system. The green checkmark confirms the non-spam classification, demonstrating the system's functionality.

	result	input
0	Ham	Go until jurong point, crazy..Available only…
1	Ham	Ok lar…Joking wifi u oni…
2	Spam	Free entry in 2 a wkly comp to win FA Cup fina…
3	Ham	U dun say so early hor… U c already then say…
4	Ham	Nah I don't think he goes to usf, he lives aro…

Table 1 : Table detecting ham or spam messages

	result	input
0	0	Go until jurong point, crazy..Available only…
1	0	Ok lar…Joking wifi u oni…
2	1	Free entry in 2 a wkly comp to win FA Cup fina…
3	0	U dun say so early hor… U c already then say…
4	0	Nah I don't think he goes to usf, he lives aro…

Table 2: Messages after preprocessing, 0 means Not spam ,1 means Spam

4.2 GitHub Link for Code:

https://github.com/Dristi-b/Spam-Detection-System

CHAPTER 5

Discussion and Conclusion

5.1 Future Work:

1. Enhancements to Data and Preprocessing:

i. Expanded Dataset: Gathering a larger and more diverse dataset, including more recent spam examples and variations in language/phrasing, will improve the model's robustness.

ii. Dynamic Data Collection: Implement a system to collect real-world SMS messages from users (with their consent) to continuously update the training data and adapt to evolving spam tactics.

iii. Advanced Preprocessing: Explore more sophisticated text preprocessing techniques like handling slang, abbreviations, misspellings, and code-mixing (using multiple languages within a single message).

iv. Sentiment Analysis: Incorporate sentiment analysis to identify the emotional tone of messages, which might be indicative of spam or phishing attempts.

2. Model Improvements:

i. Deep Learning Models: Experiment with deep learning architectures like Recurrent Neural Networks (RNNs), LSTMs, or Transformers, which are often better at capturing complex patterns in text data.

ii. Ensemble Methods: Combine multiple models (e.g., stacking or blending) to improve prediction accuracy and robustness.

iii. Feature Engineering: Develop more informative features, such as statistical features (e.g., frequency of special characters), semantic features (e.g., presence of URLs or phone numbers), or contextual features (e.g., sender information).

iv. Active Learning: Implement active learning strategies to selectively request labels for the most informative unlabeled data points, reducing the labeling effort required for model updates.

3. Deployment and User Experience:

i. Mobile App Development: Create a mobile application for seamless integration with users' messaging apps.

ii. API Integration: Develop an API to allow other applications or services to use the spam detection functionality.

iii. Real-time Updates: Implement a system for automatically retraining and updating the model with new data to maintain its effectiveness over time.

4. Addressing Emerging Threats:

i. Phishing Detection: Extend the system to detect phishing attempts via SMS, which often involve malicious links or requests for personal information.

ii. Smishing Detection: Specifically target smishing attacks, which use SMS messages to deceive users into providing sensitive information or downloading malware.

iii. Multilingual Support: Develop models that can detect spam in multiple languages to address the global nature of spam communication.

5.2 Conclusion

This system has laid the groundwork for a practical and scalable SMS spam detection solution. The deployed system effectively classifies messages in real-time, providing a valuable tool for users to manage unwanted communication. Future development will prioritize expanding the system's capabilities through multilingual support, personalized filtering, and integration with popular messaging platforms.

Moreover, continuous monitoring and retraining of the model with real-world data will be essential to maintain its accuracy and relevance in the face of ever-changing spam techniques. The ultimate goal is to create a user-friendly and highly effective spam filtering solution that empowers individuals to control their mobile communication experience.

Summary of Achievements:

i. Successfully developed and deployed a web-based SMS spam detection system.

ii. Utilized Python, Scikit-learn, Pandas, NumPy, and Streamlit.

iii. Achieved 100% precision on the selected model.

iv. Created a user-friendly interface for real-time spam classification.

v. Demonstrated the potential of machine learning for practical spam filtering.

Reflection on the Process:

i. Gained experience in data collection, preprocessing, and exploratory data analysis (EDA).
ii. Developed skills in model selection, training, and evaluation.
iii. Successfully deployed a machine learning model as a web application.
iv. Learned about the challenges and considerations involved in building a real-world spam detection system.

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